

$$1. W = \{h(f(a), g(x)), h(z, z)\}$$

$$\{f(a) = z, g(x) = z\} \quad \text{DESC}$$

$$\{z = f(a), g(x) = z\} \quad \text{Swap}$$

$$\{g(x) = f(a)\} \quad \text{Sust } [z := f(a)]$$

X

DESC Falla

$$2. W = \{f(w, f(x, h(z))), f(g(x), f(x, y)), f(g(x), f(a, b))\}$$

$$f(w, f(x, h(z))) = f(g(x), f(x, y)) \quad \text{DESC}$$

$$w = g(x), f(x, h(z)) = f(x, y) \quad \text{Sust } [w := g(x)]$$

$$f(x, h(z)) = f(x, y)$$

DESC

$$x = x, h(z) = y$$

Elim

$$h(z) = y$$

Swap

$$y = h(z)$$

Sust $[y := h(z)]$

$$f(g(x), f(x, h(z))) = f(g(x), f(a, b)) \quad \text{DESC}$$

$$g(x) = g(x), f(x, h(z)) = f(a, b) \quad \text{Elim}$$

$$f(x, h(z)) = f(a, b)$$

DESC

$$x = a, h(z) = b$$

Sust $[x := a]$

$$h(z) = b$$

DESC Falla

X

$$4. W = \{f(x, g(f(a, y), z), f(b, g(f(a, g(x, c)), f(y, x))))\}$$

$$f(x, g(f(a, y), z) = f(b, g(f(a, g(x, c)), f(y, x))) \quad \text{Desc}$$

$$x = b, g(f(a, y), z) = g(f(a, g(x, c)), f(y, x)) \quad \text{Sust} \\ [x := b]$$

$$g(f(a, y), z) = g(f(a, g(b, c)), f(y, b)) \quad \text{Desc}$$

$$f(a, y) = f(a, g(b, c)), z = f(y, b) \quad \text{Sust} \\ [z := f(y, b)]$$

$$f(a, y) = f(a, g(b, c)) \quad \text{Desc}$$

$$a = a, y = g(b, c) \quad \text{Elim}$$

$$y = g(b, c) \quad \text{Sust } [y := g(b, c)]$$

$\emptyset$

$$[x := b] [z := f(y, b)] [y := g(b, c)]$$

$$[x := b, z := f(y, b)] [y := g(b, c)]$$

$$[x := b, z := f(g(b, c), b), y := g(b, c)]$$

$$f(b, g(f(a, g(b, c)), f(g(b, c), b)))$$

$$f(b, g(f(a, g(b, c)), f(g(b, c), b)))$$